Digital Material Passport

ID **DMP-METAL-EN10029-001** - *Version* **1.0.0**
Issue Date **2025-05-20** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ThyssenKrupp Steel Europe AG Kaiser-Wilhelm-Straße 100 47166 Duisburg DE quality@thyssenkrupp-steel.example.com	**Heavy Industries Construction AB** Industrivägen 25 Göteborg 41705 SE procurement@heavy-ind.example.se	**Order** PO-2025-1547 · **24800 kg** Pos. 10 · 2025-04-10 **Delivery** DN-2025-8854 · **24800 kg** Pos. 10 · 2025-05-20

Product

Product Name	Hot-rolled structural steel plate S355J2+N	*Number (EN)*	1.0577
Batch ID	H-58742-12	*Name (Material)*	S355J2+N
Surface Condition	Hot-rolled	*Shape*	Plate - Width 2000 mm - Thickness 25 mm - Length 4500 mm
Delivery Condition	+N - Normalized	*Product Norm*	EN 10025-2 (2019)
Weight	24800 kg	*Product Norm*	EN 10029 (2010)
Production Date	2025-05-18		
Country of Origin	DE		
Tolerance Standard	EN 10029:2010		

Chemical Analysis

Heat Number **H-58742** - *Melting Process* **BOF+LF** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-17** - *Sample Location* **Ladle**

Symbol	C	Mn	Si	P	S
Unit	%	%	%	%	%
Min					
Max	0.20	1.60	0.50	0.025	0.020
Actual	0.18	1.52	0.28	0.015	0.009

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*
Tensile Strength[1]	Rm	515 MPa	470 MPa	630 MPa	EN ISO 6892-1
Yield Strength[1]	ReH	378 MPa	355 MPa		EN ISO 6892-1
Elongation after fracture At room temperature, $L0=5.65\sqrt{S0}$	A	24 %	20 %		EN ISO 6892-1
Charpy V-notch Impact Energy At -20°C, longitudinal	KV	45 J	27 J		EN ISO 148-1

[1] At room temperature

Supplementary Tests

Test	*Result / limits*	*Method*	*Status*
Thickness Measured at 4 points	24.8 mm - min 24.7 mm - max 26.3 mm	EN 10029	✓
Flatness Deviation Measuring length 2000mm, Steel type L	7 mm - max 10 mm	EN 10029	✓
Edge Camber Measured over full length	8.00 mm - max 11.25 mm	EN 10029	✓
Out-of-Squareness Measured at corners	15 mm - max 20 mm	EN 10029	✓

Validation

We hereby certify that all material described above has been manufactured and tested in accordance with the requirements of EN 10025-2:2019, EN 10029:2010, and EN 10204:2004 type 3.1. The results comply with the requirements for S355J2+N steel grade and dimensional tolerances class B-S-G.

Validated by **Dr. Werner Schmidt**, Head of Quality Control, Quality Assurance - 2025-05-20